WEEK 6 – HEALTHCARE ANALYSIS DASHBOARD (BY : GEETHASHINI.V.S)

1 . Introduction

In Week 6, an interactive Healthcare Analysis Dashboard was developed using Microsoft Power BI. The cleaned and structured healthcare data model created in Week 5 was used to analyze patient demographics, medical conditions, admissions, insurance providers and billing information. Different visualizations and interactive slicers were created to make the dashboard easy to analyze.

2 . Objectives

The main objectives of Week 6 were:

- Create DAX measures for healthcare analysis.
- Create KPI cards for important healthcare metrics.
- Analyze patients based on age group and gender.
- Analyze medical conditions and admission types.
- Analyze billing amounts by insurance provider.
- Analyze patient and billing trends over time.
- Add interactive slicers.
- Create an interactive healthcare dashboard.

3 . DAX Measures

Four measures were created for the dashboard:

Total Patients

```
Total Patients =
DISTINCTCOUNT(Billing[Patient ID])
```

Total Billing Amount

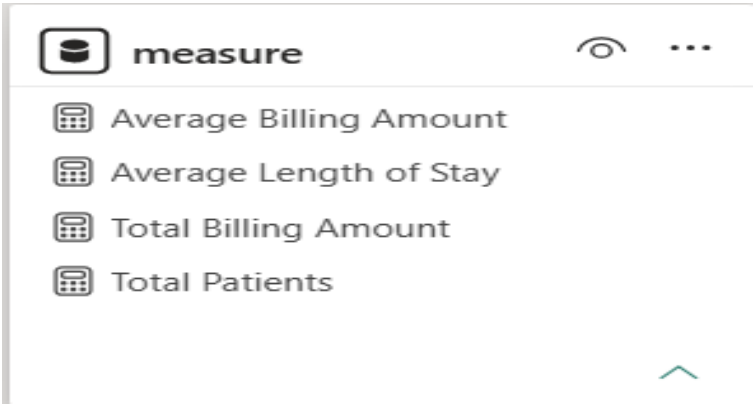
```
Total Billing Amount =
SUM(Billing[Billing Amount])
```

Average Billing Amount

```
Average Billing Amount =
AVERAGE(Billing[Billing Amount])
```

Average Length of Stay

```
Average Length of Stay =
AVERAGEX(
    Billing,
    DATEDIFF(
        Billing[Date of Admission],
        Billing[Discharge Date],
        DAY
    )
 )
```



4. KPI Summary Cards

Four KPI cards were created:

- Total Patients
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

These cards provide a quick summary of the main healthcare indicators



5. Patient Demographics – Matrix

A Matrix visual was created using:

- Rows: Age Group, Gender
- Columns: Medical Condition
- Values: Total Patients, Average Billing Amount

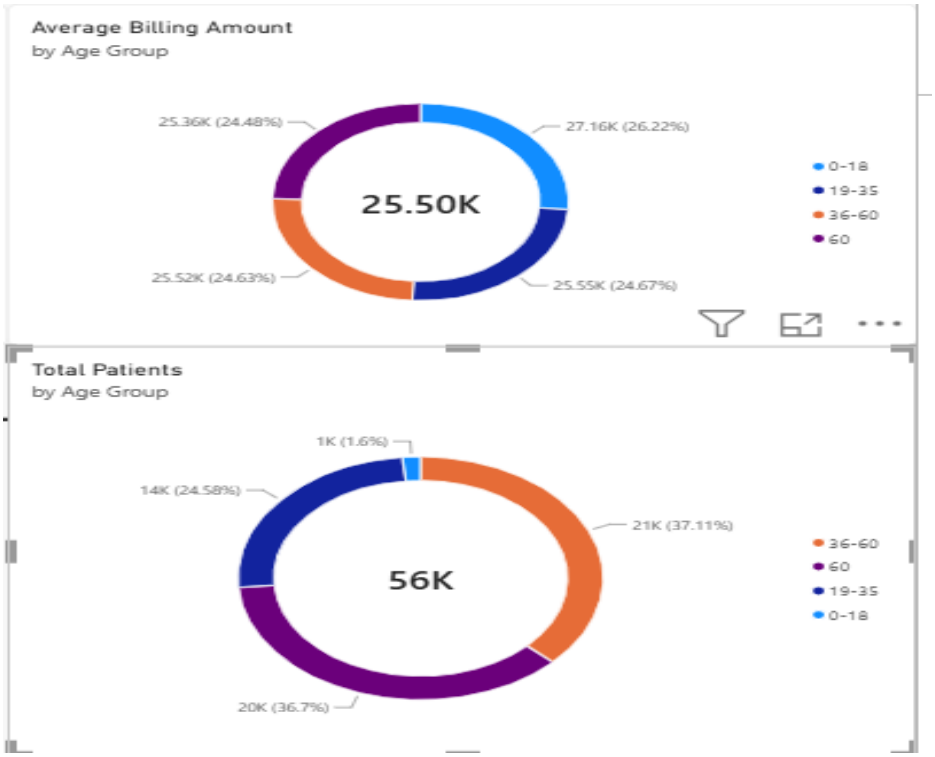
Medical Condition	Arthritis		Asthma		Cancer		Diabetes		Hypertension		Obesity		Total	
Age Group	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount
0-18	150	28,441.05	150	27,287.64	146	27,035.22	140	27,291.74	143	29,074.07	159	24,352.75	888	27,162.99
19-35	2279	25,302.80	2289	26,114.24	2288	25,389.26	2246	25,308.93	2252	25,342.13	2290	25,837.52	13644	25,554.41
36-60	3485	25,290.74	3319	25,431.03	3405	25,068.66	3499	25,530.54	3423	25,533.93	3467	26,266.85	20598	25,521.75
60	3394	25,614.62	3427	25,266.28	3388	24,817.89	3419	25,640.50	3427	25,376.20	3315	25,479.79	20370	25,364.95
Total	9308	25,467.05	9185	25,574.10	9227	25,086.35	9304	25,544.10	9245	25,482.24	9231	25,842.94	55500	25,499.47

6. Patient Distribution by Age Group

A Donut Chart was created using:

Legend: Age Group

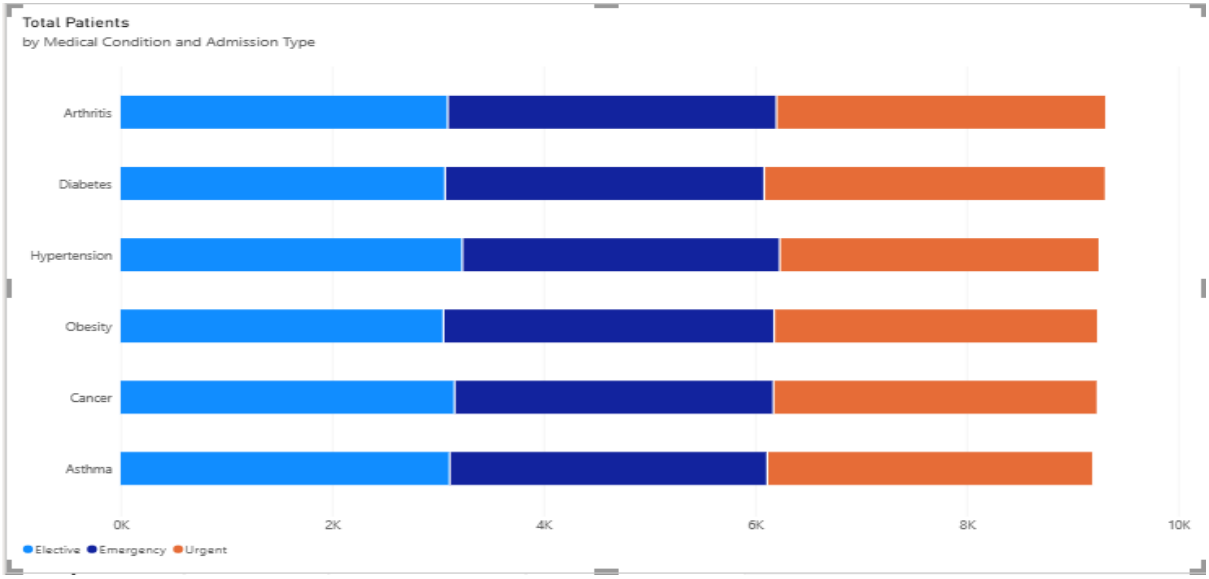
Values: Total Patients and Average Billing Amount

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7. Medical Condition and Admission Analysis

A Stacked Bar Chart was created using:

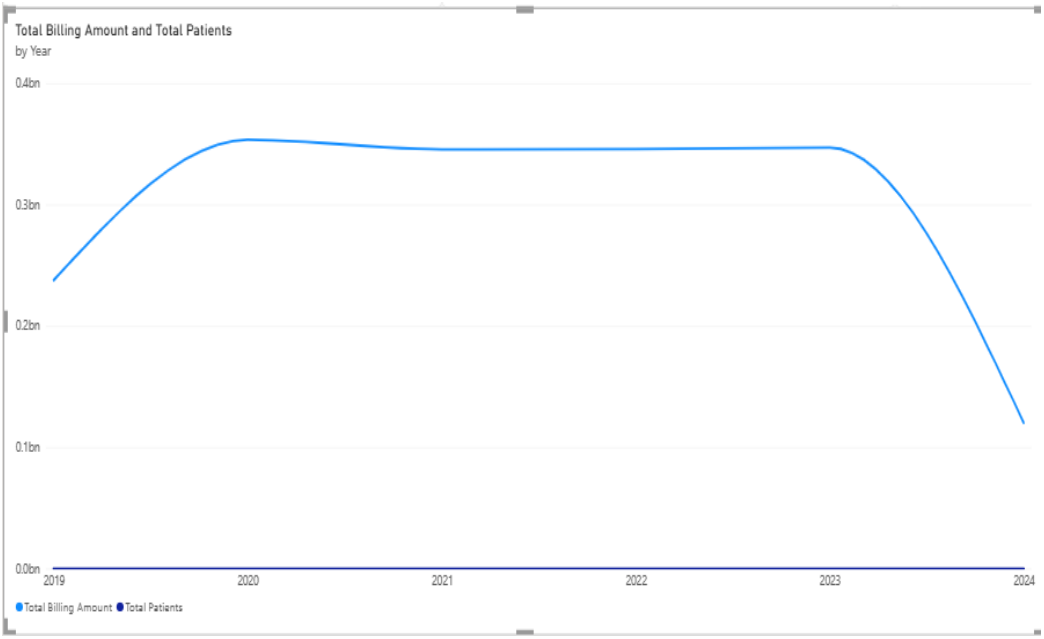
- **Y-axis:** Medical Condition
- **X-axis:** Total Patients
- **Legend:** Admission Type



8. Admission and Billing Trend

A Line Chart was created using:

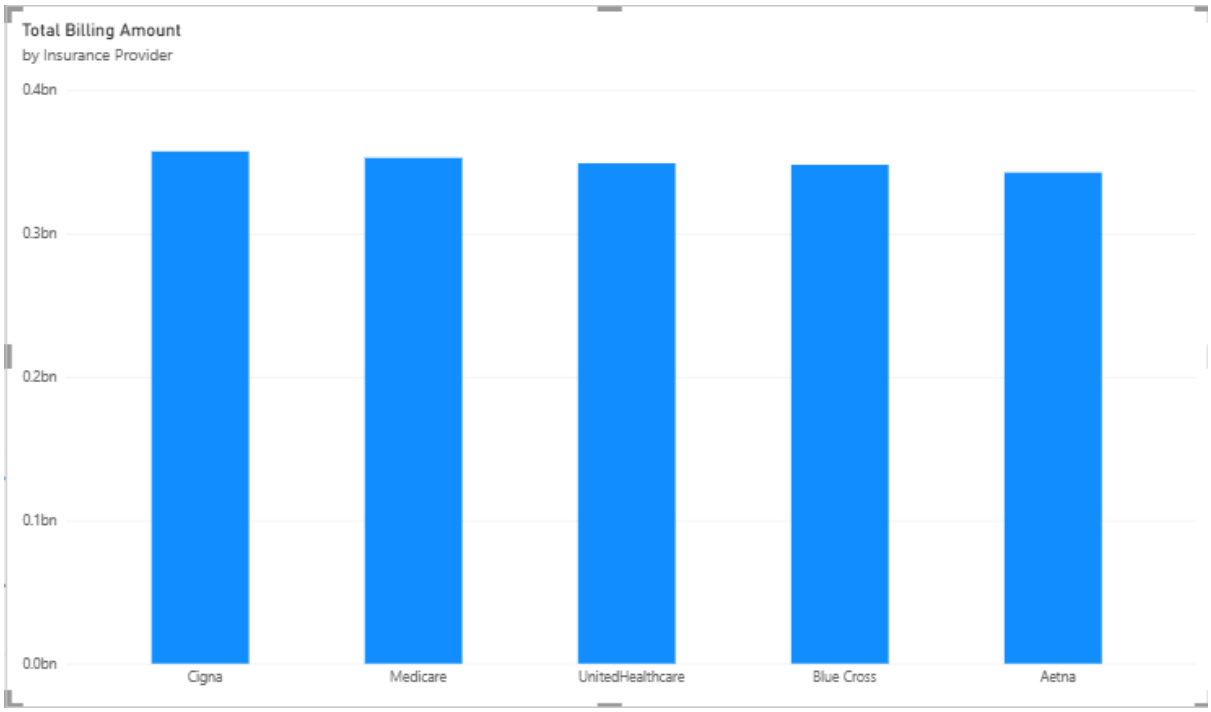
- **X-axis:** Date of Admission
- **Y-axis:** Total Patients
- **Y-axis:** Total Billing Amount



9. Insurance Provider Analysis

A Clustered Column Chart was created using:

- **X-axis:** Insurance Provider
- **Y-axis:** Total Billing Amount



10. Interactive Slicers

Three slicers were added to the dashboard:

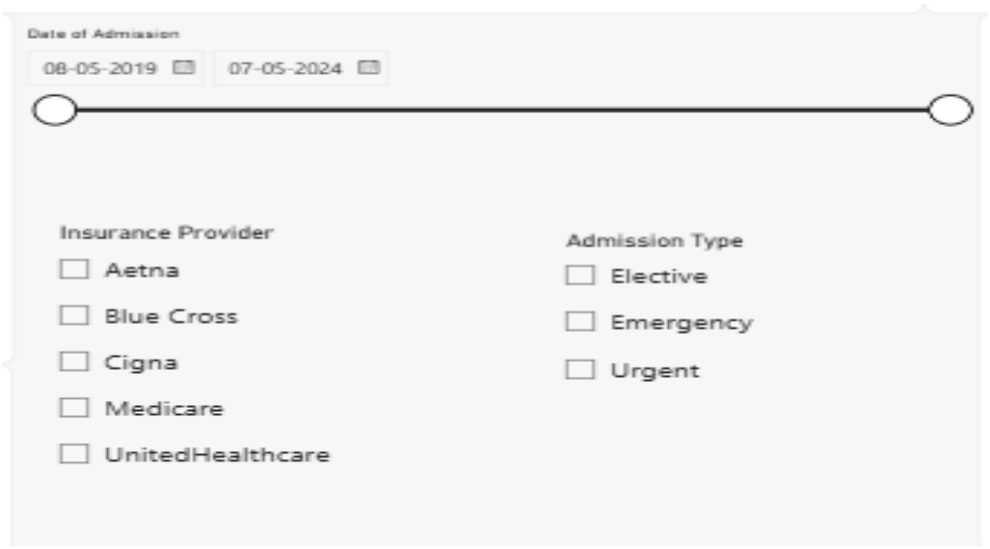
1. **Insurance Provider**
2. **Admission Type**
3. **Date of Admission**

The date slicer was configured as a **Between** date range.

11. Dynamic Cross-Filtering

The slicers were tested to ensure that the dashboard visuals respond to user selections.

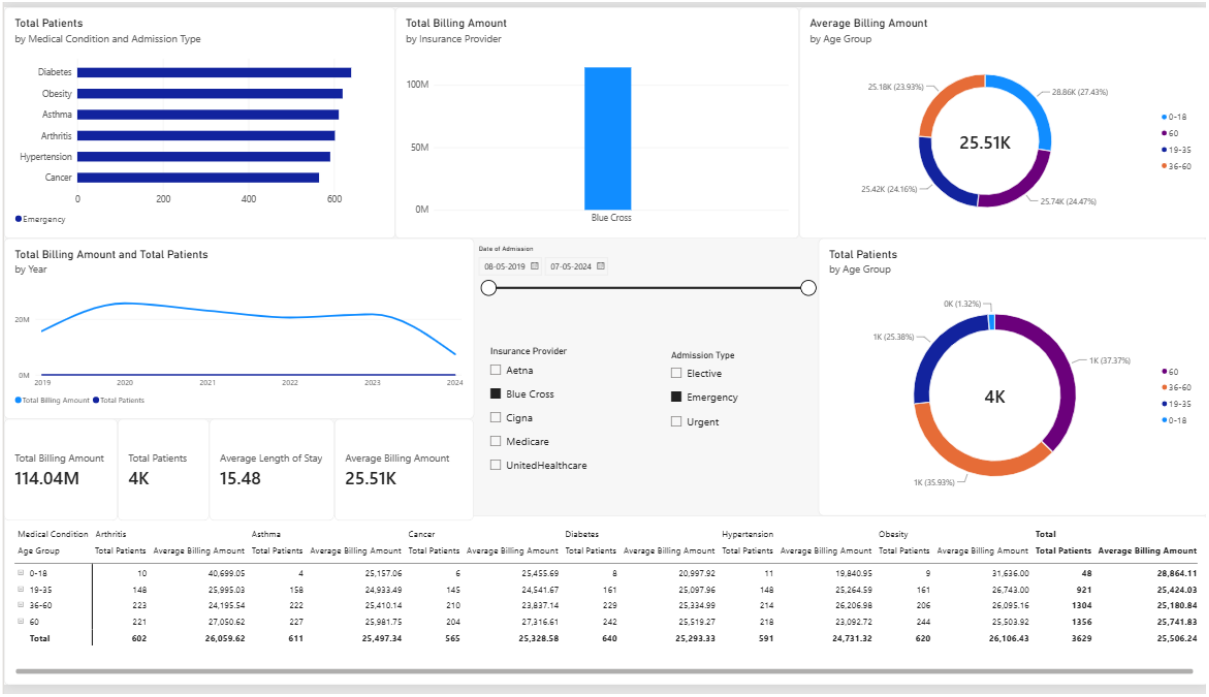
For example, selecting an insurance provider changes the values displayed in the related charts and KPI cards.



12. Final Healthcare Dashboard

All the created visuals were arranged into a single interactive dashboard.The dashboard contains:

- KPI cards
- Medical condition analysis
- Patient demographic analysis
- Age group distribution
- Admission trends
- Insurance provider analysis
- Interactive slicers



Medical Condition	Arthritis		Asthma		Cancer		Diabetes		Hypertension		Obesity		Total	
Age Group	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount	Total Patients	Average Billing Amount
0-18	10	40,699.05	4	25,157.06	6	25,455.69	8	20,997.92	11	19,840.95	9	31,636.00	**48**	**28,864.11**
19-35	148	25,995.03	158	24,933.49	145	24,541.67	161	25,097.96	148	25,264.59	161	26,743.00	**921**	**25,424.03**
36-60	223	24,195.54	222	25,410.14	210	23,837.14	229	25,334.99	214	26,206.98	206	26,095.16	**1304**	**25,180.84**
60	221	27,050.62	227	25,981.75	204	27,316.61	242	25,519.27	218	23,092.72	244	25,503.92	**1356**	**25,741.83**
Total	**602**	**26,059.62**	**611**	**25,497.34**	**565**	**25,328.58**	**640**	**25,293.33**	**591**	**24,731.32**	**620**	**26,106.43**	**3629**	**25,506.24**

13. Conclusion

In Week 6, an interactive Healthcare Analysis Dashboard was successfully developed using Microsoft Power BI. DAX measures were created to calculate important healthcare metrics, and multiple visualizations were used to analyze patient demographics, medical conditions, admissions, insurance providers and billing information. Interactive slicers were added for Insurance Provider, Admission Type and Date of Admission. The final dashboard provides a structured and interactive way to explore the healthcare dataset.